Name _____ Period _____ Date _____

5-6 Lesson Quiz: Inverse Relations and Functions

short form

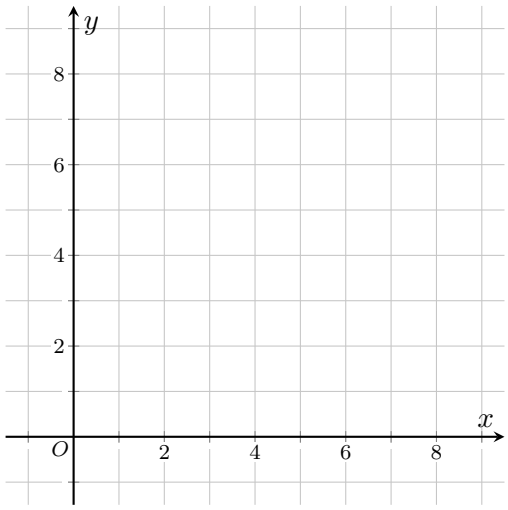
Question numbers match the Savvas Lesson Quiz (questions 1 and 3 are not on this form).

2. Sketch the graph of f and the inverse of f, if $f(x) = \frac{1}{2}x + 3$.

Show your work: switch x and y and solve for y (or swap the coordinates of two points of f).

$y = \frac{1}{2}x + 3 \longrightarrow x =$ _____

$f^{-1}(x) =$ _____



4. Write an equation of the inverse function of $f(x) = \frac{1}{2}\sqrt[3]{x-3} + 5$.

$f^{-1}(x) =$ _____ + _____ $(x -$ _____$)^3$

Show your work: switch x and y, then solve for y.

5. Let $f(x) = 2x + 4$ and $g(x) = \frac{1}{2}x - 2$. Complete the sentence. Check **one** box.

The function $g(x) = \frac{1}{2}x - 2$ _____ $f(x) = 2x + 4$.

☐ is the inverse of ☐ is equivalent to ☐ is the opposite of ☐ has the same intercepts as

Show how you know: find $(f \circ g)(x)$ and $(g \circ f)(x)$, or switch x and y in f and solve for y.

Scoring: Question 4 and Question 5 are 10 points each. Question 2 is a bonus of 5 points. Total out of 20.

5-6 Lesson Quiz, short form: answer key

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2. Graph of f and f^{-1} (bonus). f is the line through $(0, 3)$ with slope $\frac{1}{2}$ (also $(2, 4)$, $(6, 6)$). Switch and solve: $x = \frac{1}{2}y + 3$, $2(x - 3) = y$, so $f^{-1}(x) = 2x - 6$: the line through $(3, 0)$ with slope 2 (also $(4, 2)$, $(6, 6)$). The two lines are reflections across $y = x$ and meet at $(6, 6)$ on $y = x$. TE key: graph shown (two lines, shallow f with y-intercept 3, steep inverse with x-intercept 3, crossing at $(6, 6)$). Savvas quiz item 2, DOK 2, skill F65. **Bonus 5:** 2 for the graph of f, 2 for the graph of the inverse, 1 for the work shown (the equation $f^{-1}(x) = 2x - 6$, or two points of f with their coordinates swapped).

4. 3; 8; 5, that is $f^{-1}(x) = 3 + 8(x - 5)^3$. Switch: $x = \frac{1}{2}\sqrt[3]{y - 3} + 5$; $x - 5 = \frac{1}{2}\sqrt[3]{y - 3}$; $2(x - 5) = \sqrt[3]{y - 3}$; cube: $8(x - 5)^3 = y - 3$; $y = 3 + 8(x - 5)^3$. No horizontal line crosses the graph of a cube-root function twice, so the inverse is a function for all real x (Example 4B). Savvas quiz item 4, DOK 2, skill F65. **10 points:** 2 for each blank (6), 4 for the work (the switch, the cube root isolated, the cube).

5. is the inverse of. $(f \circ g)(x) = 2(\frac{1}{2}x - 2) + 4 = x - 4 + 4 = x$ and $(g \circ f)(x) = \frac{1}{2}(2x + 4) - 2 = x + 2 - 2 = x$; or $x = 2y + 4$ gives $y = \frac{1}{2}x - 2 = g(x)$. Savvas quiz item 5, DOK 2, skill F65. **10 points:** 4 for the inverse box and no other, 6 for the check (3 for each composition simplified to x, or 6 for the switch-and-solve to $\frac{1}{2}x - 2$).

Items left off this form (available for a retake or Form B): Savvas item 1, missing values in the table of the inverse (answer $a = -1$, $b = 1$, $c = 3$, $d = -2$; DOK 1, F65; taught by Example 1, cut from the fast packet; the packet's "switch the columns" line rehearses it); Savvas item 3, inverse of $f(x) = x^2 - 2x + 1$ by restricting its domain (answer B: $f^{-1}(x) = \sqrt{x} + 1$ with the domain of f restricted to $x \geq 1$; DOK 2, F65; taught by Example 3, cut from the fast packet).

IXL replacement: SmartScore 70 or above on ZRQ, 9KT and K2W replaces this quiz with 20/20.